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INT304: Network Security and Protocols – Lab 1: Introduction to Network Security Concepts

Discuss various types of network attacks, including:

- **Denial of Service (DoS) and Distributed Denial of Service (DDoS):** Flooding a network to make it unavailable. A denial-of-service attack (DoS attack) is a cyber-attack where the attacker looks for to make a machine or network resource unavailable to its deliberated users by temporarily or indefinitely services of disturbing a host connected to the Internet. Denial of service is usually accomplished by flooding the targeted machine or resource with excessive requests in an attempt to overload systems and prevent some or all legitimate requests from being fulfilled.

In a distributed denial-of-service attack (DDoS attack), the incoming traffic flooding the victim originates from many different sources. A DoS or DDoS attack is analogous to a group of people crowding the entry door or gate to a shop or business, and not letting legitimate parties enter into the shop or business, disrupting normal operations. Basically, the attacker machine either himself sends infinite request packets on the target machine without waiting for reply packet form target network or uses bots (host machines) to send request packet on the target machine. DDos attacks are used for EXTORTION, DDos as a service.

Protection against Dos and DDos attacks are:

1. Use firewall or web application firewall.
2. Use load balance or CDNS which can share traffic to different locations
3. Use DDos defense systems or service providers
4. Use a network monitoring system
5. Develop a denial of service response plan: write down the procedure that must be follow in the DDos attacks.
6. Use web filtering to block access to bad websites

- **Man-in-the-Middle (MitM):** Intercepting communication between two parties.

A man-in-the-middle (MITM) attack is a cyberattack where an attacker secretly intercepts and potentially alters the communication between two parties, allowing them to eavesdrop or impersonate one of the parties involved. This can lead to the theft of sensitive information, such as login credentials or personal data. The attacker can even inject malicious code.

- **Phishing:** Deceiving individuals into providing sensitive information. Phishing is the process of tricking a user to share sensitive personal information (PII and SPII) with the attacker, such as credit card, password etc.

The most common phishing is to send a fraudulent email that looks like it's from a trusted organization or company, when username and password enter by the user, the attacker will have access to the user information.

2. Identifying Threats and Vulnerabilities

- Conduct group exercise where each group is given a hypothetical network scenario (e.g., a small business, a university campus).
- Identify potential threats in the given scenarios, discussing what types of attacks could be launched and the vulnerabilities present in the network architecture.

Scenario:

A small business, "**Ayilara Green business enterprise**," has a network infrastructure that supports 50 employees. The network consists of:

- A router connected to the internet
- A firewall
- A switch connecting 10 departments
- 50 workstations (Windows and macOS)
- 10 mobile devices (iOS and Android)
- A server room with 5 servers (Windows and Linux)
- A guest Wi-Fi network

Potential Threats and Vulnerabilities:

1. **Phishing Attacks:** Employees may receive phishing emails or messages that trick them into revealing sensitive information or installing malware.

Vulnerability: Lack of employee training on phishing attacks and inadequate email filtering

2. **Unauthorized Access:** An unauthorized person may try to access the network through the guest Wi-Fi or by physically accessing an unsecured workstation.

Vulnerability: Weak passwords, lack of multi-factor authentication, and inadequate physical security.

3. Malware and Ransomware: Malware or ransomware may be introduced into the network through infected emails, infected software downloads, or infected mobile devices.

Vulnerability: Outdated antivirus software, lack of regular software updates, and inadequate backup procedures.

4. Denial of Service (DoS) Attacks: An attacker may launch a DoS attack on the network, overwhelming it with traffic and making it unavailable to users.

Vulnerability: Inadequate firewall configuration, lack of traffic filtering, and insufficient network bandwidth.

5. Insider Threats: A disgruntled employee may intentionally compromise the network or steal sensitive information.

Vulnerability: Lack of employee screening, inadequate access controls, and insufficient monitoring of network activity.

6. Unpatched Vulnerabilities: The network may have unpatched vulnerabilities in software or hardware, which could be exploited by an attacker.

Vulnerability: Lack of regular software updates, inadequate vulnerability scanning, and insufficient patch management.

7. Physical Security Risks: The server room and network equipment may be physically accessible to unauthorized individuals.

Vulnerability: Inadequate physical security measures, such as locks, cameras, and alarms.

Recommendations:

1. Implement employee training on phishing attacks and cybersecurity best practices.
2. Enforce strong passwords, multi-factor authentication, and regular password changes.
3. Install and regularly update antivirus software, firewalls, and intrusion detection systems.
4. Implement a backup and disaster recovery plan to ensure business continuity.
5. Conduct regular vulnerability scanning and patch management.
6. Implement physical security measures, such as locks, cameras, and alarms, to protect the server room and network equipment.

7. Develop an incident response plan to quickly respond to security incidents.

3. Understanding Security Protocols

Review key security protocols and their functions:

SSL/TLS (Secure Sockets Layer/Transport Layer Security) operates by establishing a secure, encrypted connection between a client (like a browser) and a server. It encrypts the data to ensure that sensitive information such as personal data and payment information remains private during transmission. TLS is the modern successor to SSL, providing enhanced security features and improved performance. It is commonly used in web browsing (HTTPS), email, and other secure data transmission applications.

IPsec (Internet Protocol Security) secures IP communications by providing both encryption and authentication at the IP layer. It operates by encapsulating and encrypting data packets for secure transfer between devices over an IP network. IPsec is typically used in virtual private networks (VPNs) and other secure network environments where data integrity and confidentiality are crucial.

SSH (Secure Shell) works by allowing secure remote login and command execution over unsecured networks. SSH encrypts the data transferred, preventing eavesdropping, connection hijacking, and other attacks. It is widely used by system administrators to access and manage servers remotely.

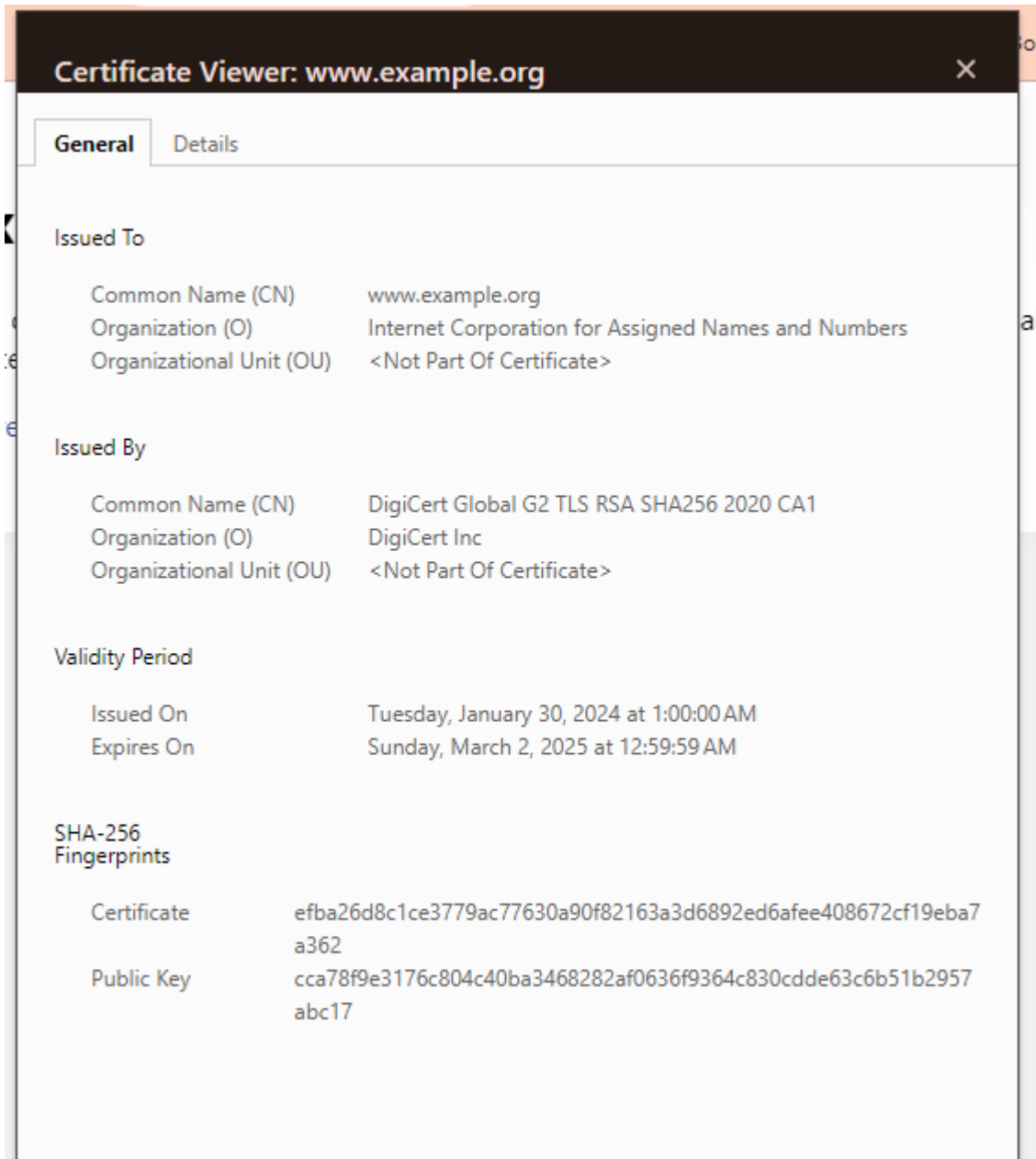
HTTPS (Hypertext Transfer Protocol Secure) is an extension of HTTP that utilizes SSL/TLS to secure communications over networks. It encrypts the data exchanged between web browsers and websites, ensuring that data remains confidential and is not tampered with. It is primarily used for securing sensitive transactions on websites, such as online banking, shopping, and login forms.

4. Hands-On Activity

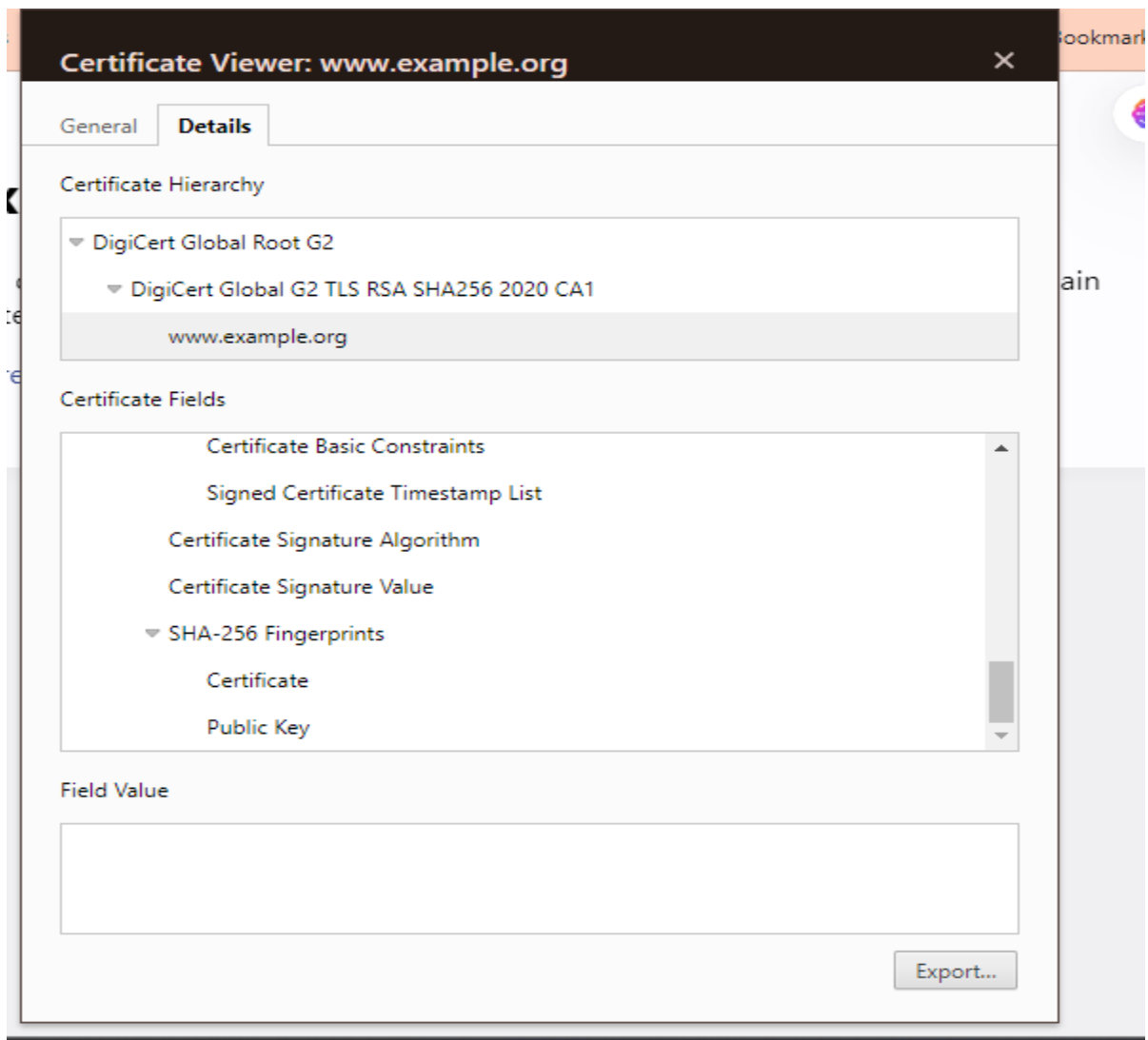
1. Open a web browser and navigate to a secure website (e.g., <https://www.example.com>).

2. Click on the padlock icon in the address bar to view the SSL/TLS certificate details.

▪ **Record the certificate issuer, validity period, and encryption strength used for the connection.**



The certificate issuer is DigiCert Inc, validity period is Sunday, March 2, 2025 at 12:59:59Am, and encryption strength: SHA-256



3. Discuss findings with your peers about the significance of having secure connections with everyday internet use.

Having secure connections in everyday internet use is crucial for protecting personal information and preventing unauthorized access. With the rise of cybersecurity threats, including data breaches and identity theft, ensuring that your internet connection is secure is essential. Using secure connections, such as those provided by VPNs or HTTPS websites, can significantly reduce the risk of interception by malicious actors, especially when using public Wi-Fi, which is often less secure. Following best practices for internet safety helps safeguard sensitive data and enhances overall online privacy.

INT304: Network Security and Protocols – Lab 2: Network Security Policies and Risk Management

1. Introduction to Network Security Policies

- Start with a reading assignment on network security policies and their significance in safeguarding organizational assets.
- Discuss the key components of an effective security policy, including:
 - **Purpose and Scope:** What the policy aims to protect.
 - **User Responsibilities:** Expectations and acceptable use of network resources.
 - **Incident Response Plan:** Steps to take when a security breach occurs.
 - **Compliance:** Adherence to relevant laws and regulations.

2. Conducting a Risk Assessment

- Break into small groups and select a hypothetical organization (e.g., a healthcare provider, a financial institution).
- Each group should identify potential threats and vulnerabilities specific to their chosen organization.
- Conduct a risk assessment using the following steps:
 - **Asset Identification:** List key assets that require protection (e.g., customer data, intellectual property).
 - **Threat Identification:** Identify potential threats to each asset (e.g., cyberattacks, natural disasters).
 - **Vulnerability Assessment:** Analyze weaknesses that could be exploited by threats.
 - **Impact Analysis:** Evaluate the potential impact of each threat on the organization.
 - **Risk Rating:** Assign a risk rating (high, medium, low) based on likelihood and impact.

3. Developing Security Policies

- Using the findings from the risk assessment, each group will draft a preliminary security policy tailored to their organization. Key elements to include:
 - **Access Control Policies:** Guidelines for user access and authentication.
 - **Data Protection Policies:** Measures to safeguard sensitive information.
 - **Incident Response Procedures:** Steps to be taken in the event of a security incident.
 - **Training and Awareness Programs:** Plans for educating employees about security best practices.

4. Presentation of Policies

- Each group will present their risk assessment and proposed security policy to the class.
- Encourage peer feedback and discussion on the strengths and weaknesses of each policy.

Scenario: Healthcare Provider

organization Name: Kwasu Teaching Hospital

Asset Identification:

1. Customer data such as sensitive patient's information, medical record, and other confidential data
2. Intellectual property such as Knowledge and Expertise, Hospital employees possess specialized knowledge and expertise that are critical to the delivery of quality patient care.
3. Medical device that connects to the internet or use for virtual operation in the Lab or theater

Potential Threats identification:

1. **Data Breach:** Unauthorized access to patient records, medical history, and financial information.
2. **Ransomware Attack:** Malware attack that encrypts hospital data, demanding payment in exchange for the decryption key.
3. **Medical Device Hacking:** Hackers gaining access to medical devices, such as insulin pumps or pacemakers, to manipulate or disrupt their function.
4. **Phishing Attacks:** Employees receiving phishing emails or messages that trick them into revealing sensitive information or installing malware.
5. **Operational Continuity:** Employees play a crucial role in maintaining the hospital's operational continuity, including ensuring that medical equipment is functioning properly, and that patients receive timely and effective care.
6. **Reputation and Trust:** Employees are often the face of the hospital, interacting with patients, families, and the community. Their behavior and performance can impact the hospital's reputation and trust.
7. **Insider Threats:** Employees with authorized access to sensitive information and systems can intentionally or unintentionally compromise security.

8. Human Error: Employees can make mistakes, such as misconfiguring medical devices or incorrectly administering medication.

Vulnerabilities Assessment:

1. Outdated Software: Legacy medical devices and software that are no longer supported or updated, leaving them vulnerable to exploitation.

2. Insufficient Employee Training: Lack of employee training on cybersecurity best practices, making them more susceptible to phishing attacks and other social engineering tactics.

3. Unsecured Wi-Fi Networks: Unsecured or poorly configured Wi-Fi networks that allow unauthorized access to the hospital's network.

4. Inadequate Incident Response Plan: Lack of a comprehensive incident response plan, making it difficult for the hospital to respond quickly and effectively in the event of a security incident.

Security Policies

1. Access Control Policies

- Access Controls: Implement strict access controls, including role-based access control, to ensure that employees only have access to the information and systems they need to perform their jobs.
- Employee Background Checks: Conduct thorough background checks on all employees, including contractors and vendors.

2. Data Protection Policies

- Monitor user activities on your system using tools like SIEM to know who accesses data at point in time.
- Follow the least privileged principle.
- Use network segmentation to isolate critical assets from the rest of the network.
- Use of strong password and Multi factor authentication (MFA)
- Revoke third-party access immediately when no longer needed.

3. Incident Response Procedures

- Incident Response Planning: Develop and regularly test incident response plans to ensure that employees know how to respond in the event of a security incident.

4. Training and Awareness Programs

- Security Awareness Training: Provide regular security awareness training to employees, including training on phishing, social engineering, and other cyber threats.

